Reasoning test

For admission to Class 8

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 7 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 7.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, using ideas taught up to Class 7. Chapters are from the NCERT (CBSE) books a student joining Class 8 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Ganita Prakash

- *Patterns in Mathematics* – number patterns and series
- *Perimeter and Area*
- *Prime Time* – primes, factors, HCF and LCM
- *Fractions*

Class 7 · Ganita Prakash

- *Working with Fractions* – fractions
- *Finding Common Ground* – primes, factors, HCF and LCM
- *Finding the Unknown* – simple equations

Class 7 · Curiosity (science)

- *Measurement of Time and Motion* – speed, distance and time

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Curiosity

- *Exploring Magnets* – magnets
- *A Journey through States of Water* – evaporation and condensation

Class 7 · Curiosity

- *Exploring Substances: Acidic, Basic and Neutral* – acids, bases and indicators
- *Electricity: Circuits and their Components* – electric circuits
- *Changes around Us: Physical and Chemical* – physical and chemical changes, rusting
- *Heat Transfer in Nature* – heat and conductors
- *Measurement of Time and Motion* – speed
- *Life Processes in Animals* – respiration
- *Life Processes in Plants* – photosynthesis and plant respiration
- *Light: Shadows and Reflections* – light, shadows and plane mirrors

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Class 7 syllabus questions

Five maths and five science questions come from these Class 7 chapters only: the first-term chapters, because the test can be taken before the year ends.

Maths

Class 7 · Ganita Prakash

- *Ch 1. Large Numbers Around Us* – lakhs and crores, place value
- *Ch 2. Arithmetic Expressions* – order of operations, brackets
- *Ch 3. A Peek Beyond the Point* – decimals
- *Ch 4. Expressions using Letter-Numbers* – algebraic expressions
- *Ch 5. Parallel and Intersecting Lines* – angles made by lines
- *Ch 6. Number Play* – even and odd, number patterns
- *Ch 7. A Tale of Three Intersecting Lines* – triangles

Science

Class 7 · Curiosity

- *Ch 2. Exploring Substances: Acidic, Basic and Neutral* – acids, bases, indicators, neutralisation
- *Ch 3. Electricity: Circuits and their Components* – cells, switches, conductors, circuits
- *Ch 4. The World of Metals and Non-metals* – properties and reactions of metals
- *Ch 5. Changes around Us: Physical and Chemical* – physical and chemical changes, rusting

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Find the next number in the series: 7, 10, 16, 25, 37, ___

   A 49 **B** 52
   C 55 **D** 61

2. ENGLISH · BASICS
   The old bridge was too fragile to carry the loaded truck. As used here, "fragile" most nearly means

   A ancient **B** narrow
   C weak **D** crowded

3. SCIENCE · BASICS
   Which of these will a magnet attract?

   A a copper wire **B** an iron pin
   C aluminium foil **D** a glass bead

4. MATHS · CLASS 7
   How many thousands make one lakh?

   Type-in answer: _____

5. ENGLISH · BASICS
   Choose the correct word: Each of the players ___ given a new jersey.

   A were **B** was
   C are **D** have

6. SCIENCE · CLASS 7
   Which of these turns red litmus paper blue?

   A Fresh lemon juice **B** White vinegar
   C Baking soda solution **D** Sour orange juice

7. MATHS · BASICS
   What is the largest number that divides both 48 and 72 exactly?

   A 24 **B** 36
   C 12 **D** 6

8. ENGLISH · BASICS
   Pen is to writer as brush is to ___

   A painter **B** canvas
   C colour **D** wall

9. SCIENCE · BASICS
   Which of these tastes sour because it contains an acid?

   A vinegar **B** soap solution
   C baking soda solution **D** chalk powder

10. MATHS · CLASS 7
    What is the value of 20 − 12 ÷ 4?

    A 2 **B** 17
    C 8 **D** 23

11. ENGLISH · BASICS
Choose the word most nearly opposite in meaning to "ancient".

A modern
B old
C large
D famous

12. SCIENCE · CLASS 7
Which of these elements is a non-metal?

A Iron
B Sulphur
C Copper
D Aluminium

13. MATHS · BASICS
A square garden has a perimeter of 48 m. What is its area, in square metres?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: She is very fond ___ music.

A with
B of
C for
D at

15. SCIENCE · BASICS
A ball rolls 45 m across flat ground in 9 seconds at a steady speed. What is its speed, in metres per second?

Type-in answer: _____

16. MATHS · CLASS 7
Which decimal is equal to the fraction 3/4?

A 0.34
B 3.4
C 0.75
D 0.43

17. ENGLISH · BASICS
Bird is to nest as bee is to ___

A honey
B flower
C hive
D wing

18. SCIENCE · CLASS 7
Which of these changes produces a new substance?

A Tearing paper
B Dissolving salt
C Burning wood
D Melting wax

19. MATHS · BASICS
Riya cycles to school at 12 km/h and reaches in 25 minutes. How far is her school from home?

A 3 km
B 4 km
C 5 km
D 6 km

20. ENGLISH · BASICS
What comes next in the series? AZ, BY, CX, DW, ___

A DV
B EU
C FX
D EV

21. SCIENCE · BASICS
Two identical bulbs are connected in parallel to a battery and both glow. If one bulb is unscrewed, the other bulb

A goes out
B glows dimmer
C stays lit as before
D flickers and fails

22. MATHS · CLASS 7

Two angles together make a straight line. One is three times the other. What is the smaller angle, in degrees?

Type-in answer: _____

23. ENGLISH · BASICS

Although the path was steep, the cadets ___ the summit before noon.

A avoided
B lost
C feared
D reached

24. SCIENCE · CLASS 7

A circuit has a cell, a bulb and a switch. The switch is closed but the bulb does not glow. What is the most likely cause?

A The bulb's filament is broken
B The switch is in the on position
C The wires are made of copper
D The cell has just been bought

25. MATHS · BASICS

Asha spends one third of her pocket money on books and one quarter of what is left on snacks. She has ₹150 left. How much pocket money did she have?

A ₹200
B ₹250
C ₹300
D ₹360

26. ENGLISH · BASICS

Some pilots are engineers. All engineers are graduates. Which statement must be true?

A All pilots are graduates
B No pilot is a graduate
C Some pilots are graduates
D All graduates are pilots

27. SCIENCE · BASICS

A leaf on a plant is partly covered with black paper for two days, then tested with iodine. Only the uncovered part turns blue-black. This shows that

A iodine reacts with the black paper
B the covered part lost all its water
C the leaf breathes only where it is uncovered
D starch is made only where light reaches

28. MATHS · CLASS 7

If n is an odd number, which of these is always even?

A n + 2
B 3n
C n + 5
D 2n + 1

29. ENGLISH · BASICS

In a row of children, Asha is 7th from the left and 12th from the right. How many children are in the row?

A 17
B 18
C 19
D 20

30. SCIENCE · CLASS 7

Antacid tablets relieve acidity in the stomach because they

A are acidic and add acid
B kill the germs in the stomach
C are sweet and soothe pain
D are basic and neutralise acid

Answers

1	**B. 52**	The gaps grow by 3 each time: +3, +6, +9, +12, so the next gap is +15, and 37 + 15 = 52.
2	**C. weak**	Fragile means easily broken. The sentence says the bridge could not carry the weight, which points to weakness, not age or width.
3	**B. an iron pin**	Magnets attract iron and a few other metals. Copper and aluminium are metals but are not attracted, and glass is not a metal.
4	**100**	One lakh is 1,00,000 and one thousand is 1,000, so a lakh is 100 thousands. This is a type-in question.
5	**B. was**	The subject is "each", which is singular, so the verb is "was". "Players" sits nearby but is not the subject.
6	**C. Baking soda solution**	Bases turn red litmus blue, and baking soda solution is basic. Lemon juice, vinegar and orange juice are acidic.
7	**A. 24**	48 = 24 × 2 and 72 = 24 × 3, and 2 and 3 share no common factor, so 24 is the largest. 36 does not divide 48.
8	**A. painter**	A pen is the tool a writer works with; a brush is the tool a painter works with. The other options are things a brush touches.
9	**A. vinegar**	Vinegar contains acetic acid, which gives it a sour taste. Soap and baking soda are bases, and chalk is neither sour nor acidic.
10	**B. 17**	Division comes before subtraction: 12 ÷ 4 = 3, and 20 − 3 = 17. Working left to right gives (20 − 12) ÷ 4 = 2.
11	**A. modern**	Ancient means belonging to the distant past, so its opposite is modern. "Old" means nearly the same as ancient.
12	**B. Sulphur**	Sulphur is a brittle yellow solid that does not conduct electricity: a non-metal. Iron, copper and aluminium are metals.
13	**144**	A square has four equal sides, so each side is 48 ÷ 4 = 12 m, and the area is 12 × 12 = 144 square metres. This is a type-in question.
14	**B. of**	"Fond" is followed by "of": fond of music, fond of reading. The other prepositions are not used with fond.
15	**5**	Speed is distance divided by time: 45 m ÷ 9 s = 5 m/s. This is a type-in question: the answer is typed as a number.
16	**C. 0.75**	3/4 = 75/100 = 0.75. 0.34 just writes the digits 3 and 4 after the point, which is the usual mistake.
17	**C. hive**	A nest is where a bird lives; a hive is where a bee lives. Honey is what bees make and flowers are where they feed.
18	**C. Burning wood**	Burning turns wood into ash, smoke and gases, new substances that cannot become wood again. The others change only shape or state.

19	**C. 5 km**	At 12 km/h she covers 1 km every 5 minutes, so in 25 minutes she covers 5 km. The minutes have to be turned into hours first.
20	**D. EV**	The first letter moves forward one step (A, B, C, D, E) while the second moves back one step (Z, Y, X, W, V), giving EV.
21	**C. stays lit as before**	In a parallel circuit each bulb has its own path to the battery, so removing one leaves the other's path complete and unchanged.
22	**45**	Angles on a straight line add to 180°: x + 3x = 180, so x = 45° (and the other is 135°). This is a type-in question.
23	**D. reached**	"Although" signals a contrast with the difficulty: despite the steep path, they succeeded. Only "reached" completes that contrast.
24	**A. The bulb's filament is broken**	A broken filament leaves a gap in the circuit, so no current flows. A closed switch, copper wires and a new cell all help the bulb glow.
25	**C. ₹300**	After books, 2/3 is left; snacks take a quarter of that, leaving 3/4 of 2/3, which is 1/2. Half is ₹150, so she had ₹300. ₹360 comes from taking a quarter of the whole.
26	**C. Some pilots are graduates**	The pilots who are engineers must be graduates, so at least some pilots are graduates. Nothing is said about the other pilots.
27	**D. starch is made only where light reaches**	Iodine turns blue-black where there is starch. Only the part that received light made starch, so the leaf needs light to make food.
28	**C. n + 5**	Odd + odd is even, and 5 is odd, so n + 5 is always even. n + 2 and 3n stay odd, and 2n + 1 is odd for every n.
29	**B. 18**	Counting from both ends counts Asha twice, so the row has 7 + 12 − 1 = 18 children. Adding without subtracting gives 19.
30	**D. are basic and neutralise acid**	Too much stomach acid causes acidity. An antacid is a mild base, such as milk of magnesia, which neutralises the extra acid.